

Project Report (35% Completion Stage)

"From Waste to Growth: Can RO Reject Water Grow Bacteria?"

1. Abstract

Domestic reverse osmosis (RO) purifiers generate 3–4 litres of reject water for every litre of purified water. This reject water is typically drained as waste despite being mineral-rich and salty. The present study investigates whether domestic RO reject water can serve as a free, alternative growth medium for halotolerant bacteria and whether the grown bacteria can produce useful bioproducts such as biofertilizers, pigments, and biofilms. At this stage of the project (approximately 35% completion), water samples have been collected from 3–5 households and physicochemical characterization (pH and TDS) has been performed. The results confirm that RO reject water has significantly higher TDS and slightly alkaline pH compared to tap water. This characterization lays the foundation for the next phase: bacterial isolation and cultivation in RO reject water.

1. Introduction

2.1 Background

Home RO purifiers are widely used for drinking water purification. However, for every 1 glass of clean water produced, 3–4 glasses are rejected as "reject water." This water contains high concentrations of dissolved salts, minerals, and sometimes trace organic matter. It is usually discarded into drains, contributing to water wastage and environmental salinity.

2.2 Problem Statement

RO reject water is treated as a pollution problem and a waste disposal issue. No one considers it a potential resource. Meanwhile, microbiological growth media and bioproduct production remain expensive due to the cost of synthetic chemicals.

2.3 Hypothesis

Domestic RO reject water can serve as a free, nutrient-rich growth medium for salt-tolerant (halotolerant) bacteria, and these bacteria may produce useful bioproducts such as biofertilizers, pigments, and biofilms.

2.4 Novelty

|What others do / What this project does |

Study RO reject water as a pollution problem	Study it as a free food source for bacteria
Use industrial brine from factories	Use domestic RO reject water (unexplored)
Focus on filtering or disposal	Focus on growing useful microbes in it
Need expensive lab chemicals	Use the waste water itself as the medium

****Novelty statement:**** *Domestic RO reject water is an unexplored, free, and salty growth medium. This project checks whether salt-tolerant bacteria can grow in it and produce useful bioproducts turning a waste disposal problem into a resource.*

1. Aim and Objectives

3.1 Aim

To evaluate the potential of domestic RO reject water as a growth medium for halotolerant bacteria and to screen the isolates for useful bioproducts.

3.2 Objectives

1. To collect and characterize RO reject water samples from domestic sources.
2. To isolate and cultivate bacteria from RO reject water.
3. To assess bacterial growth in RO reject water supplemented with glucose.
4. To perform preliminary screening tests for biofertilizer activity, pigment production, and biofilm formation.
5. To compare bacterial growth in RO reject water versus normal tap water.

6. Materials and Methods

4.1 Materials

| Item |
| RO reject water | (3–5 sources) |
| Tap water |
| TDS meter |
| pH paper |
| Nutrient agar plates |
| Glucose |
| Test tubes, inoculating loop |
| Incubator |
| Crystal violet |
| Sterile filter paper / cotton |
| Seeds (mung bean / wheat) |
| Ethanol (70%) |
| Bleach / autoclave |

4.2 Methods

4.2.1 Sample Collection

RO reject water was collected from 3–5 homes in clean, sterile bottles. Tap water was collected as a comparison. All samples were stored at 4°C until use.

4.2.2 Physicochemical Characterization

- **pH:** Measured using pH paper.
- **TDS:** Measured using a TDS meter.

4.2.3 Isolation of Bacteria *(Planned — Not Yet Started)*

- A drop of RO reject water will be spread on nutrient agar plates.
- Plates will be incubated at 37°C for 24–48 hours.
- Colony morphology (shape, color, size) will be recorded.

4.2.4 Cultivation in RO Reject Water *(Planned — Not Yet Started)*

- Single colonies will be picked with a sterile loop.
- Inoculated into test tubes containing RO reject water + 1% glucose.
- Incubated at 37°C for 24–48 hours with shaking.
- Growth will be assessed by turbidity (cloudiness).

4.2.5 Biofertilizer Test (Seed Germination Bioassay) *(Planned — Not Yet Started)*

- Two sets of sterile Petri dishes will be prepared with sterile filter paper.
- **Test plates:** soaked with 1–2 mL bacterial culture in RO reject water.
- **Control plates:** soaked with 1–2 mL sterile RO reject water (no bacteria).
- 5–10 surface-sterilized seeds (mung bean/wheat) will be placed in each plate.
- Incubated at 25–30°C for 3–5 days.
- Germination percentage, root length, and shoot length will be measured.

4.2.6 Biofilm Test (Crystal Violet Assay) *(Planned — Not Yet Started)*

- Test tubes with RO reject water broth will be inoculated and incubated stationary at 37°C for 24–48 hours.
- Liquid will be poured out and tubes gently rinsed with water.
- 0.1% crystal violet solution will be added and left for 10–15 minutes.
- Tubes will be rinsed gently until water runs clear.
- A purple ring/film on tube walls will indicate a positive result.

4.2.7 Pigment Test *(Planned — Not Yet Started)*

- Liquid cultures will be observed visually for color change after 24–48 hours.
- Colony color on agar plates will also be recorded.
- If pigment is intracellular, cells will be centrifuged, re-suspended in ethanol/methanol, vortexed, and centrifuged again. The supernatant containing pigment will be observed.

Observations:

- RO reject water showed **significantly higher TDS** (780–920 ppm) compared to tap water (150 ppm).
- The pH of RO reject water was **slightly alkaline** (7.7–7.9) compared to tap water (7.0).
- These results confirm that RO reject water is **mineral-rich and salty**, as expected.

5.2 Work Yet to Be Completed

The following work is planned but not yet started:

- Bacterial isolation on nutrient agar plates
- Cultivation in RO reject water broth
- Biofertilizer test (seed germination bioassay)
- Biofilm test (crystal violet assay)
- Pigment test
- Comparative analysis of growth in reject water vs tap water

1. Discussion (Based on Work Completed)

The physicochemical characterization confirms that RO reject water has significantly higher TDS than tap water. This is consistent with the fact that RO systems concentrate salts and minerals in the reject stream. The slightly alkaline pH (7.7–7.9) is also typical of RO reject water.

These characteristics — high TDS, alkaline pH, and mineral richness — make RO reject water a potentially suitable environment for halotolerant bacteria. While high salinity can inhibit many ordinary bacteria, salt-loving (halotolerant) bacteria can thrive in such conditions. This is the basis of our hypothesis.

The next phase of the project will test whether bacteria can actually grow in this water and whether they produce useful bioproducts. The TDS and pH data provide a baseline for comparing growth in reject water versus tap water.

It is important to note that some bacteria that may grow in RO systems can be opportunistic pathogens (e.g., **Pseudomonas aeruginosa**, **Aeromonas caviae**). Therefore, all future culturing work will be handled with strict hygiene, and no identification to species level will be performed without proper safety protocols.

1. Conclusion (Interim)

At this 35% completion stage, the project has successfully demonstrated that:

1. RO reject water can be collected from domestic sources.
2. It has significantly higher TDS and slightly alkaline pH compared to tap water.
3. These characteristics make it a potentially suitable medium for halotolerant bacteria.

The remaining work includes bacterial isolation, cultivation, and bioproduct screening tests. The preliminary characterization supports the hypothesis that RO reject water is not waste — it is a potential resource.

1. Future Scope (Remaining Work — 65%)

2. Isolation of bacteria from RO reject water on nutrient agar plates.
3. Cultivation of isolates in RO reject water broth supplemented with glucose.

4. Biofertilizer test (seed germination bioassay).
5. Biofilm test (crystal violet assay).
6. Pigment test.
7. Comparison of bacterial growth in RO reject water vs tap water.
8. Identification of bacterial isolates using 16S rRNA sequencing.
9. Optimization of growth conditions (pH, temperature, glucose concentration).
10. Quantitative analysis of biofertilizer activity (IAA, phosphate solubilization).
11. Extraction and characterization of pigments (UV-Vis spectrophotometry, TLC).
12. Scale-up for bioproduct production using RO reject water as a cheap medium.
13. Safety assessment of isolates for environmental and industrial applications.

Appendix: Safety Notes

- All future cultures will be treated as potentially pathogenic.
- Gloves and lab coats will be worn at all times.
- Work surfaces will be disinfected with 70% ethanol before and after work.
- All waste will be autoclaved or bleached before disposal.
- Plates and tubes will never be opened and sniffed.